

Coarse-Graining and the Classification of Long-Range Correlations in Quantum Field Theory

Jinku Guo ^a

^a*Northwestern Polytechnical University, Xi'an 710072, China*

E-mail: guojk@nwpu.edu.cn

ABSTRACT: A conservation-law-guided classification framework is developed for the execution of coarse-graining operations in quantum field theory. Coarse-graining produces a selection at the level of Feynman diagrams. Diagrams with nonzero momentum transfer are suppressed by oscillatory factors, zero-momentum-transfer ladder diagrams can accumulate through geometric series resummation to produce a spectral pole, and single-bubble topologies contribute only to the continuum. Conservation laws govern this classification. For operators protected by a Ward identity, the matrix element at zero momentum is guaranteed to be nonvanishing. For operators protected by BRST symmetry, the Slavnov-Taylor identities provide no mandatory suppression. The two cases differ in the strength of the algebraic guarantee. For unprotected operators the injection term vanishes and the spectral function remains continuous. The sign of the single-bubble contribution is determined by spin statistics. A positive sign leads to amplificative feedback in the ladder resummation, a negative sign leads to suppressive feedback. These two attributes, the nonvanishing of the injection term and the sign of the single-bubble contribution, are the defining criteria of the classification. As a direct application, a general classification of local operators is established within the emergence framework and verified on eight physical channels and three known solvable systems. The framework indicates which emergence paths are possible for each operator; whether the critical condition is reached is left for independent nonperturbative computation. The logical structure of this classification is parallel to the strategy used in deriving fluid equations from molecular kinetic theory in classical statistical physics.

KEYWORDS: Coarse-graining; Conservation laws; Ward identity; Spectral function; Operator classification; Emergent gravity; Functional renormalisation group

Contents

1	Introduction	2
2	Three Criteria of Coarse-Graining	4
2.1	What Coarse-Graining Does at the Level of Feynman Diagrams	4
2.2	How Conservation Laws Guarantee a Nonvanishing Injection Term	7
2.3	Reliability of the Ladder Approximation	9
2.4	Systematic Execution Steps	10
3	Operator Classification	12
3.1	Basic Principles	12
3.2	Symmetry Origin of the Criteria	15
3.3	Protected Operators	16
3.4	Unprotected Operators	16
3.5	Scheme Selection	17
3.6	Physical Consequences and Falsifiability	18
3.7	A Hierarchical Interpretation	19
4	Irreversibility of Coarse-Graining and the RG Time	20
4.1	Irreversibility of Coarse-Graining	20
4.2	Scalar Condensation Driven by the RG Flow	21
4.3	The Emergence Window as an RG Time Window	21
5	Synergy between Analytic Classification and Numerical Computation	22
5.1	Verification on Eight Physical Channels	23
5.2	Cross-Validation Logic	26
6	Discussion and Summary	27
A	Generalised Källén–Lehmann Spectral Representation	30
A.1	Generalised spectral representation	30
A.2	Monotonicity of the window function	30
A.3	Transmission of the zero-frequency divergence	31
A.4	Hermiticity restriction	31
A.5	Scope of applicability	32
B	Operator Hierarchy from the CGC Classification	32
B.1	Motivation	32
B.2	Protection and Self-Driving	32
B.3	Four Levels	33
B.4	Structure Diagram	34
B.5	Consequences	34

1 Introduction

How microscopic degrees of freedom produce macroscopic collective behaviour through coarse-graining is a question that runs through statistical physics, fluid dynamics, condensed matter physics and quantum field theory. In classical statistical physics the rigorous derivation of the Navier–Stokes equations from Boltzmann’s molecular kinetics has been completed in recent work, which established the macroscopic fluid equations as the inevitable consequence of microscopic molecular collisions in the thermodynamic limit [1]. In cosmology the effective field theory of large-scale structure shows how symmetry principles and coarse-graining govern the nonlinear clustering of dark matter. Short-range physics is absorbed into a finite set of transport coefficients, yielding high-precision predictions for the observed galaxy distribution [2]. In condensed matter physics the topological order of the fractional quantum Hall effect, the pseudogap phase of high-temperature superconductors and non-Fermi-liquid behaviour all involve long-range correlations whose structure is constrained by conservation laws [3]. In quantum gravity, Sakharov’s induced gravity, Jacobson’s thermodynamic gravity and Verlinde’s entropic force proposal have approached gravity as an emergent phenomenon from the perspectives of quantum fluctuations, thermodynamic relations and the holographic principle respectively [4–6]. More recently the development of generalised global symmetries has provided a formal framework in which symmetry protection constrains the correlation functions of operators at a fundamental level [7].

What these developments share is a common logical structure. The emergence of long-range correlations is constrained by conservation laws and the kinematic structure of coarse-graining, while microscopic details affect only non-universal transport coefficients. This logic has not yet been systematically implemented in the nonperturbative sector of quantum field theory. The main tools currently available for nonperturbative problems have each achieved important successes but share a structural limitation.

Lattice gauge theory discretises spacetime onto a finite lattice and evaluates the path integral by Monte Carlo methods. It has been successful for the QCD mass spectrum, hadron structure and quark confinement, but it is constrained by fermion doubling, the sign problem at finite density and computational cost. Lattice results by themselves do not reveal which diagrammatic channels are responsible for the structures seen in spectral functions. The functional renormalisation group follows the evolution of the effective action with the coarse-graining scale through the Wetterich equation. It can track spectral functions, effective potentials and critical behaviour, and is widely applied to critical phenomena, quantum gravity and strongly correlated systems. It does not, however, automatically identify which diagram classes accumulate under coarse-graining and which are suppressed, and its results depend on the choice of regulator. Recent advances in spectral functional renormalisation group methods [8] have made it possible to compute spectral functions of quantum field theories directly, though the regulator dependence and

the lack of an analytic classification remain. Holographic duality maps strongly coupled gauge theories to weakly coupled gravitational theories via the AdS/CFT correspondence, providing direct access to the strong coupling regime. It applies, however, only to systems that possess a gravitational dual and is not directly applicable to the Standard Model itself. Dyson–Schwinger equations start from the quantum equations of motion and solve for Green’s functions with nonperturbative truncations. They can handle mass generation and chiral symmetry breaking but depend on the choice of truncation scheme.

The common limitation of these four tools is that they yield quantitative results without supplying an analytic criterion that determines, before any numerical computation, which operators and which interaction channels possess the potential to develop long-range correlations. This judgement ordinarily relies on complete numerical computations whose outcome itself depends on the regulator scheme, computational resources and model choices.

The idea that conservation laws select physically admissible subsets of Feynman diagrams has a precedent in perturbative gauge theory. Boos and Ohl constructed minimal gauge-invariant classes of tree-level diagrams, known as groves, by requiring that a subset of diagrams satisfy the Ward identities by themselves [9]. The classification developed here applies the same underlying logic of conservation laws as a selection criterion to a different physical setting. Instead of ensuring gauge invariance at tree level, the criterion determines which operators can develop long-range correlations under the nonperturbative operation of coarse-graining.

In a previous work [10] the author established coarse-graining as a dynamical principle for emergent gravity and proposed a spectral criterion, namely whether the spin-2 spectral density of the energy–momentum tensor commutator develops an isolated pole at zero momentum. Within the spectral-flow class considered there, a smooth spectral function is not a fixed point of the coarse-graining flow; the isolated delta pole is the selected non-smooth fixed-point configuration, and Weinberg’s low-energy theorem locks the resulting low-energy effective theory uniquely to the Einstein–Hilbert form. The present paper takes the next step and gives the diagrammatic execution of the coarse-graining operation, the operator-level classification, and the link between the classification criterion and the Standard Model dynamical channels. A second work [11] tested the operator distinction implied by the emergence framework in the electroweak sector. Taking the ultraviolet boundary condition $\lambda(M_P) = 0$ for the unprotected Higgs quartic coupling, two-loop renormalisation group evolution yielded a Higgs mass consistent with the measured value within theoretical uncertainties.

The present paper develops a conservation-law-guided classification framework that supplies an analytic selection mechanism. Within the stated class of local operators, without relying on perturbative expansions or the strength of couplings, and using only conservation laws and the kinematic structure of Feynman diagrams, it determines which operators can develop long-range correlations under coarse-graining. The framework makes the coarse-graining operation of Ref. [10] precise at the diagrammatic level. Whether the momentum transfer vanishes separates diagrams that can accumulate from those that are suppressed by oscillatory factors. Whether the diagram topology is a ladder separates

diagrams capable of producing a spectral pole from those that merely contribute to the continuum. Whether the operator is protected by a conservation law separates operators with a nonvanishing injection term from those whose injection term vanishes. These criteria introduce no new physical hypotheses. They systematise, algorithmise and make general the diagram classification ideas contained in Appendix E of Ref. [10].

The classification logic is structurally similar to the Landau paradigm in condensed matter [12]. In that framework, symmetry dictates which excitations can emerge at low energies. Here symmetry plays the same role. Ward identities and BRST invariance determine which operators can accumulate spectral weight under coarse-graining. The CGC framework is the QFT analogue of the Landau classification for long-range correlations [13].

The execution of coarse-graining involves several interrelated aspects. At the diagrammatic level it is necessary to clarify which Feynman diagrams make cumulative contributions to the slow-mode effective action after fast modes are integrated out, which are suppressed by fast-mode oscillations, and whether the selection criteria depend on the order of perturbation theory or the size of couplings. At the operator level different operators display sharply different behaviour under coarse-graining. Whether an operator can generate long-range correlations, and why, is traceable to the conservation properties of the operator and the spin statistics of the fields to which it couples. A further distinction is drawn between dynamical ladder accumulation and topological zero-mode mechanisms. The framework analyses the former; the latter, where relevant, is identified as a separate pathway. The classification framework developed here systematically answers these two questions. It is orthogonal to the standard RG classification by scaling dimension; it addresses pole accumulation rather than operator growth under scale change. A complementary verification branch cross-checks the analytic classification against nonperturbative numerical computations. The classification framework identifies the available emergence paths. The FRG computation directly evaluates the propagator and the pole condition. Together the analytic classification and the numerical verification form a complete chain from analytic discrimination to dynamical confirmation. CGC supplies the structural selection; the FRG supplies the dynamical realisation. The framework also distinguishes situations where the ladder pathway is closed by topology, as with the scalar glueball G^2 . Such channels belong to the topological branch rather than to the dynamical ladder branch.

2 Three Criteria of Coarse-Graining

2.1 What Coarse-Graining Does at the Level of Feynman Diagrams

The coarse-graining flow of the spectral density derived in Ref. [10] consists of two terms. A dilution term suppresses spectral weight at all nonzero momenta through the exponential window. An injection term deposits the spectral weight of the integrated fast modes back into the surviving slow modes. The two terms are not independent. Their combined effect is equivalent to a set of selection rules operating directly on the Feynman diagrams that constitute the spectral function. The dilution term governs the suppression of diagrams whose fast internal lines carry a momentum transfer that oscillates under the

coarse-graining window. The injection term governs the redistribution of spectral weight through diagrams whose topology permits a cumulative build up over successive coarse-graining steps. The present section makes this correspondence precise. The two criteria developed below, one kinematic and one graph theoretic, map the dilution and injection terms onto explicit diagrammatic selection rules.

Coarse-graining integrates out the fast modes with an exponential window function $W_\sigma(p) = \exp(-p^2\sigma^2/2)$. This operation is irreversible. In quantum field theory the fast and slow modes are coupled by interactions, and their degrees of freedom are correlated in the path integral. Integrating out the fast modes amounts to taking the partial trace over the fast Hilbert space. The reduced density matrix of the slow modes is obtained. Because the entanglement between the fast and slow modes is removed, the reduced density matrix turns from a pure state into a mixed state, and its von Neumann entropy increases. At the level of Feynman diagrams the effect of fast-mode integration is characterised by two criteria. The two criteria are independent of each other, respectively kinematic and graph-theoretic, and do not rely on the specific form of the interactions or on the strength of the couplings.

The first criterion concerns whether the momentum transfer vanishes. Consider a Feynman diagram containing internal fast lines and external slow legs. The net momentum transfer q from the fast internal lines to the slow external legs is a purely kinematic quantity, determined by the conservation of energy and momentum at each vertex. When $q \neq 0$ the integration over the fast modes produces an oscillatory factor $e^{iq \cdot x}$. In the infrared limit of the coarse-graining flow the coarse-graining scale σ becomes macroscopic, and this oscillatory factor is exponentially suppressed by the integration over spacetime within the coarse-graining window, vanishing in the limit. Diagrams with $q \neq 0$ therefore give a negligible net contribution in the infrared. They do not participate in the formation of a spectral pole but instead feed the noise kernel $\eta(t, x)$ in the Langevin equation (2.4) of the order parameter. When $q = 0$ the fast modes enter the vertices that connect to the slow external legs in a back-to-back configuration, meaning that the fast momenta appear in pairs of $+k$ and $-k$, with zero net momentum transfer. This is the characteristic momentum routing of ladder diagrams. In this case there is no oscillatory factor, and the contribution of the diagram can accumulate over successive coarse-graining steps. This classification is a standard result of Fourier analysis and is mathematically rigorous. It does not depend on the order of perturbation theory or on the size of the couplings. It depends only on the momentum routing of the diagram, and the momentum routing is itself fixed solely by energy-momentum conservation.

The second criterion concerns diagram topology. Among the diagrams with $q = 0$ a further distinction must be made according to topological structure. The simplest are single-bubble diagrams, which contain a single closed fast loop and no irreducible internal insertions. The contribution of a single-bubble diagram to the spectral function remains smooth in the infrared limit, producing only a continuum $\rho_{\text{cont}}(s)$ and no isolated pole. Another important class consists of ladder diagrams. A ladder diagram is built from several independent bubbles connected in a chain by irreducible vertices V . The irreducible vertex V does not itself contain fast propagators. It is an effective four-operator interaction among

the slow modes, generated by the second-order effect of integrating out the fast modes at an earlier coarse-graining step. The characteristic feature of ladder diagrams is that the fast momenta inside each bubble retain the back-to-back configuration, so that every individual bubble satisfies the $q = 0$ condition. The contribution of the entire ladder can be summed as a geometric series (2.1)

$$\Pi_{\text{resum}} = \Pi_0 + \Pi_0 \cdot V \cdot \Pi_0 + \Pi_0 \cdot V \cdot \Pi_0 \cdot V \cdot \Pi_0 + \dots = \frac{\Pi_0}{1 - V \cdot \Pi_0}, \quad (2.1)$$

where Π_0 is the single-bubble contribution at zero momentum. The closed form of this geometric series shows that when the denominator tends to zero the spectral function develops an isolated δ pole at $p^2 = 0$. The critical condition is $1 - V \cdot \Pi_0 = 0$.

The above critical condition takes into account only the bare polarisation Π_0 . In the actual nonperturbative dynamics the dressing of the propagator feeds back onto Π_0 itself. The dressed polarisation is enhanced relative to the bare one because each internal line in the bubble receives its own dressing factor, producing the relation $\Pi_{\text{dressed}} = \Pi_0 / (1 - V \cdot \Pi_{\text{dressed}})^2$. From this one obtains the cubic bifurcation equation (2.2)

$$y(1 - y)^2 = V \cdot \Pi_0, \quad (2.2)$$

with $y = V \cdot \Pi_{\text{dressed}}$ (substituting $y = V \Pi_{\text{dressed}}$ into $\Pi_{\text{dressed}} = \Pi_0 / (1 - V \Pi_{\text{dressed}})^2$ and multiplying by V). The physical branch and the unstable branch of this equation merge at $y = 1/3$, where $y(1 - y)^2$ attains its maximum, corresponding to the critical value

$$V \cdot \Pi_0 = \frac{4}{27} \approx 0.148. \quad (2.3)$$

The critical value (2.3) is reached earlier than the direct zero of the geometric series denominator $V \cdot \Pi_0 = 1$ and is the physically correct threshold for emergence.

The momentum transfer criterion and the topology criterion address two distinct questions. The first determines which diagrams survive the oscillatory suppression of coarse-graining. The second identifies, among the survivors, the specific class of diagrams capable of producing a spectral pole through geometric series resummation. Neither criterion, however, addresses a further question. The selection rules derived so far assume that the vertices coupling the fast shell modes to the slow external legs have nonvanishing matrix elements at zero momentum transfer. If those matrix elements were to vanish, no amount of kinematic survival or topological accumulation would produce an injection term, and no spectral weight would flow. Whether the relevant matrix elements are nonvanishing is decided by a property that lies outside the kinematic and graph theoretic analysis. This property is the protection of the operator by a conservation law. The analysis of this third criterion follows.

Together the three criteria form a complete decision tree. The momentum transfer classification removes the diagrams that cannot accumulate. The topology classification singles out the ladder diagrams that can produce a pole. The conservation law check decides whether those ladder diagrams have nonvanishing matrix elements, thereby determining whether an injection term exists to drive the accumulation.

2.2 How Conservation Laws Guarantee a Nonvanishing Injection Term

Two complementary arguments are given. One is based on the dynamical mechanism of critical slowing down under coarse-graining, the other on a rigorous proof using the algebraic properties of the Noether current. The two arguments lead independently to the same conclusion.

2.2.1 Dynamical mechanism

Reference [10] established coarse-graining as a dynamical principle and derived the Langevin dynamics of the order parameter. The coarse-graining scale $\sigma(x)$ is promoted to a spacetime-dependent order parameter field whose Langevin equation reads

$$\partial_t \sigma = \frac{\sigma}{c} (D \nabla^2 \sigma + \beta(\sigma) + \eta), \quad (2.4)$$

with $\beta(\sigma) = -\delta V_{\text{eff}}/\delta \sigma$ the deterministic drift and η the stochastic noise generated by the fast modes. The effective temperature at the coarse-graining scale σ is $k_B T_{\text{eff}} = \hbar c/2\pi\sigma$, whose physical origin lies in the equivalence between the coarse-graining operation and a local acceleration [10]. At the infrared fixed point σ_c the critical slowing down condition $\beta'(\sigma_c) = 0$ eliminates the restoring force on the order parameter and its relaxation time diverges. The fluctuations $\delta\sigma$ therefore develop a divergent correlation function at zero frequency. This zero-frequency divergence is transmitted to the spectral function of an operator through the monotonic coupling of the window function. The coarse-grained spectral density is defined as $\rho_F(p^2; \sigma) = W_\sigma(p) \rho_F(p^2)$. The dependence of the window function on σ is strictly monotonic, $\partial W_\sigma/\partial \sigma = -p^2 \sigma \cdot W_\sigma < 0$ for all $p^2 > 0$. A fluctuation $\delta\sigma$ induces a fluctuation of the spectral density through the chain rule, $\delta \rho_F = (\partial W_\sigma/\partial \sigma) \delta\sigma \cdot \rho_F$. Because the derivative is strictly negative, the zero-frequency divergence is transmitted from $\delta\sigma$ to $\delta \rho_F$ without suppression, causing the spectral weight to diverge at $p^2 = 0$.

The premise for this transmission is that the bare spectral function of the operator contains no cutoff-dependent divergences. If the bare spectral function still contains uncanceled power-law divergences or logarithmic divergences, these cutoff-dependent terms absorb the zero-frequency divergence through the same coupling, leaving the spectral function continuous at zero momentum. Whether an operator is protected by a conservation law therefore directly determines whether the zero-frequency divergence can reach the spectral function freely. For protected operators the Ward identities have already eliminated all power-law divergences from physical correlation functions, and the zero-frequency divergence becomes the only singularity that cannot be cancelled. For unprotected operators the surviving cutoff-dependent divergences absorb the zero-frequency divergence.

2.2.2 Algebraic argument

The injection term Δ_σ in the coarse-graining flow is the spectral weight deposited into the slow modes after the fast modes are integrated out. At the perturbative level the nonvanishing of the injection term relies on the Seeley–DeWitt coefficient a_2 of the heat kernel expansion [14], which assumes that nonperturbative effects do not cancel this coefficient. The present work establishes the nonvanishing of the injection term as a rigorous consequence of conservation laws, without relying on perturbative estimates.

For the energy–momentum tensor $T_{\mu\nu}$, the conserved Noether current of spacetime translations, a standard corollary of the Ward identity states [15] that the matrix element of a conserved Noether current J_μ at zero momentum equals the matrix element of the conserved charge Q between the same initial and final states,

$$\langle f|J_\mu(0)|i\rangle\big|_{q=0} \propto \langle f|Q|i\rangle. \quad (2.5)$$

For the energy–momentum tensor the conserved charge is the total energy–momentum P_μ . Any particle state that carries energy and momentum satisfies $\langle p|P_\mu|p\rangle = p_\mu \neq 0$. Hence the matrix element of the energy–momentum tensor at zero momentum is nonvanishing as an algebraic fact, independent of the order of perturbation theory, the size of the couplings, or the details of nonperturbative corrections.

The gauge field strength $F_{\mu\nu}^a$ is protected by BRST symmetry. It is BRST closed, $s_B F = 0$, and its matrix elements are protected by the Slavnov–Taylor identities. BRST closure guarantees that the correlation functions of gauge-invariant operators do not depend on the gauge fixing. For the two-point function of $F_{\mu\nu}^a$ the Slavnov–Taylor identity does not force the matrix element to vanish at $q = 0$. The structural logic is parallel to that of the energy–momentum tensor. In both cases the conservation law eliminates the mechanism that would otherwise force the matrix element to vanish at zero momentum. The protected status of an operator consists precisely in this elimination. For $T_{\mu\nu}$, the conserved charge on single particle states supplies a direct argument for a nonvanishing matrix element. For the gauge field strength no strictly analogous charge argument is available. Its protected status rests on the same structural ground as that of the energy–momentum tensor. The Slavnov–Taylor identity provides no mandatory suppression at zero momentum transfer. Both operators belong to the protected class, though the proof of nonvanishing is stronger for the energy–momentum tensor.

The injection term is regenerated at every step of the coarse-graining process. Because it originates from the shell of fast modes that is integrated out at the current coarse-graining scale, and every such shell contains the same matrix elements of the protected operator, the injection term is nonvanishing at every step. The injection term therefore acts as a nonvanishing source that continuously deposits spectral weight throughout the entire coarse-graining flow, not merely at some particular scale.

For protected operators the matrix elements of the zero-momentum-transfer ladder diagrams are nonvanishing, and the injection term continuously injects spectral weight. For unprotected operators the injection term vanishes, and the spectral function remains continuous in the infrared limit. The injection term is nonvanishing not because perturbation theory suggests it might be, but because the algebraic properties of the conserved Noether current require it to be. The result is no longer an estimate. It follows from the algebra.

The dynamical mechanism and the algebraic argument, though logically independent, converge to the same physical conclusion for complementary reasons. The dynamical mechanism reveals the physical origin of the nonvanishing injection term. Critical slowing down generates a zero-frequency singularity that cannot be absorbed for protected operators, because all divergences that could absorb it have already been eliminated by the Ward identities. The algebraic argument supplies a proof for specific operators that does not

depend on the details of the critical slowing down dynamics. For the energy–momentum tensor, the algebraic argument provides a rigorous demonstration of the nonvanishing based on the relation between the Noether current and the conserved charge. For the gauge field strength, the algebraic argument establishes that the Ward identity provides no mandatory suppression at zero momentum, placing the operator in the same protected class. The two arguments are not redundant. The first explains why the classification works, and the second proves that it must.

The fluctuations of an ordinary composite operator are gradually diluted by coarse-graining and eventually fade away. The fluctuations of the energy–momentum tensor are different. Their spectral weight is protected by the Ward identity. It cannot disappear. It can only be redistributed across momentum space. It is this forced redistribution that makes it possible for spectral weight ultimately to condense into an isolated pole at zero momentum.

2.3 Reliability of the Ladder Approximation

The ladder approximation is a tool for the quantitative resummation step. The classification conclusions do not depend on it. The nonvanishing of the injection term and the sign of Π_0 follow from algebraic and group-theoretic arguments that do not involve any resummation topology. The FRG computation determines the critical scale. It solves the full Wetterich equation and automatically includes all diagrammatic topologies. It does not require a separate completeness proof for the ladder approximation.

The following three arguments provide supporting evidence for the ladder approximation as a reasonable starting point for the FRG computation.

2.3.1 Guarantee from conservation laws

The conservation laws provide a guarantee that does not rely on the ladder approximation. The nonvanishing of the injection term is an algebraic consequence of the conservation laws. Even if non-ladder diagrams produce additional contributions, they can only shift the quantitative position of the critical condition, not the qualitative conclusion that the injection term is nonvanishing. Whether an operator possesses emergence potential does not require the ladder approximation to be valid. No matter whether crossed diagrams, vertex corrections or other multi-loop topologies are suppressed, the Ward identity or BRST symmetry guarantees the nonvanishing of the injection term. If the ladder approximation were quantitatively imprecise, that would merely correct the critical scale, without altering the judgement that the operator has emergence potential.

2.3.2 Spectral sparseness of the emergence window

The spectral sparseness of the emergence window supplies an independent physical basis for the ladder approximation. When the coarse-graining window contains only a single effectively active field mode, the topological distinction between crossed diagrams and ladder diagrams loses its meaning. A crossed diagram requires at least two independent active modes to build its crossed momentum routing. When only one mode is active, the topological structures of crossed and ladder diagrams degenerate into different representations of

the same diagram. The independent contributions of vertex correction diagrams and other multi-loop topologies are likewise restricted to a single momentum configuration. Any diagram built on a single active mode has its possible structures severely compressed. The ladder approximation is therefore not an approximation in this regime. It is exact when the mode count is exactly one, and highly accurate when the active modes form a small, sparse set. This conclusion does not rely on a large- N expansion or on any suppression parameter, but is a rigid consequence of counting the modes of the internal space that lie within the emergence window.

Whether this spectral sparseness condition is realised depends on the spectrum of the internal space at the relevant coarse-graining scale. In the regime in which the emergence mechanism operates, the internal space is highly curved and its nonzero spectral gaps are large compared with the coarse-graining scale. With the Litim regulator [16], a sharp step function used widely in FRG practice for its optimal convergence in the derivative expansion, only modes whose eigenvalue lies below k^2 contribute to the trace density [17]. Under these conditions the only mode that satisfies this condition is the scalar $J = 0$ zero mode, whose eigenvalue is zero and therefore always lies below the cutoff, together with the nearby modes that remain within the window. Using the exponential window $\exp(-p^2\sigma^2/2)$ of Ref. [10] instead gives every mode an exponentially small contribution rather than a sharp cutoff, but the scalar zero mode continues to dominate under any reasonable regulator. The compact-spectrum branch of the framework performs the quantitative mode counting for the specific internal space; the present argument uses only the verified sparseness of the active mode set.

A completeness theorem covering all possible non-ladder diagram classes is addressable within the FRG, where the functional resummation automatically captures all diagram topologies. The current verification covers the known relevant classes, and the crossed and ladder topologies have been shown to degenerate on a sparse spectrum. The FRG treatment subsumes the completeness question: the critical scale is determined by the full Wetterich equation, which is not restricted to any particular diagram class.

2.3.3 Colour factor suppression

Colour factor suppression operates in the multi-mode region. At the equilibrium configuration, where the number of active modes is about nine, the ratio of crossed diagrams to ladder diagrams lies between roughly 0.1 and 0.5. In the perturbative regime the group theory of $SU(3)$ gives a colour factor suppression for crossed ladders, $R = 1/(2C_A^2) \approx 1/18$. Emergence, however, takes place inside the window where $N_{\text{eff}} = 1$, not in the multi-mode region. The colour factor suppression in the perturbative regime is auxiliary. It shows that even in regions without the protection of spectral sparseness the contribution of crossed diagrams remains at a moderate level of suppression.

2.4 Systematic Execution Steps

The conservation-law-guided classification framework is implemented as a six-step pipeline.

(1) Given a target composite operator, all connected irreducible Feynman diagrams are generated at the one-loop level. The generation can be performed using the built-in

field type enumerator, which automatically determines the coupled field types from the conservation property of the operator and then produces for each field type a $q = 0$ bubble and a $q \neq 0$ variant, or it can call an external tool such as QGRAF [18] for cross-checking. For the four core operators of the emergence framework the expected numbers of diagrams are six for the energy–momentum tensor, four for the gauge field strength, two for the fermion bilinear and four for the Higgs quartic coupling.

(2) Each generated diagram is classified by its net momentum transfer q from the fast internal lines to the slow external legs. Diagrams with $q = 0$ proceed to the next step; diagrams with $q \neq 0$ are labelled as suppressed and feed the Langevin noise kernel. This is a purely kinematic judgement that does not depend on the specific form of the interactions. For the energy–momentum tensor one expects three $q = 0$ diagrams and three $q \neq 0$ diagrams; for the gauge field strength, two of each; for the fermion bilinear, one of each; for the Higgs quartic coupling, two of each.

(3) The $q = 0$ diagrams are sorted by topology into three classes. Single-bubble diagrams, which contain only one closed loop without any irreducible insertion, are labelled as the continuum class and contribute only to the continuum. Ladder diagrams, in which several bubbles are connected in a chain by irreducible vertices, are labelled as the accumulation class and can produce a δ pole through geometric series resummation. Non-standard topologies such as crossed ladders and vertex corrections are labelled as the pending class. At the one-loop level all $q = 0$ diagrams are single-bubble diagrams; ladder diagrams are multi-loop structures generated automatically by the resummation module.

(4) A conservation law check determines whether the ladder diagrams in the accumulation class have nonvanishing matrix elements. Protected operators, such as the energy–momentum tensor (protected by the Ward identity) and the gauge field strength (protected by BRST symmetry), have a nonvanishing injection term and therefore possess emergence potential. Unprotected operators, such as the fermion bilinear and the Higgs quartic coupling, have a vanishing injection term and no emergence potential.

(5) For the accumulation class diagrams that have passed the conservation law check, the geometric series resummation is performed and the critical condition $1 - \lambda_{\text{eff}}\Pi_0(0) = 0$, equivalently $V \cdot \Pi_0 = 4/27$, is output. Whether the effective coupling indeed reaches the critical value is a dynamical question that must be answered by FRG or DSE.

(6) The results are cross-validated against an independent FRG computation. The classification results are transmitted to the FRG engine through a standardised JSON interface. The FRG engine solves the complete Wetterich equation on the discrete spectrum of a compact internal space and returns the spectral function and the pole information. The cross-validation module compares the outputs of the two independent computational paths, namely the classification path and the direct continuum path, and produces a consistency report on the sign of Π_0 , the nonvanishing of the injection term and the existence of a pole. The pipeline runs automatically from operator specification to final classification verdict.

3 Operator Classification

The distinct critical behaviour exhibited by the energy–momentum tensor and the Higgs quartic coupling under coarse-graining has its origin in symmetry. The former is the conserved Noether current of spacetime translations. Its vacuum correlation functions are protected by Ward identities, power-law divergences cancel automatically within commutators, and the zero-frequency divergence produced by critical slowing-down propagates freely to the spectral function, accumulating as an isolated pole at zero momentum. The latter has no such protection. Its logarithmic divergence survives in physical correlation functions, the zero-frequency divergence is absorbed, and the spectral function stays continuous at the critical point.

A local operator is called a protected operator if its vacuum matrix elements are protected by conservation laws or Ward identities, so that cutoff-dependent divergences cancel automatically in physical correlation functions and infrared observables are independent of the ultraviolet regularisation scheme. A local operator is called an unprotected operator if its vacuum expectation values enjoy no such protection, are sensitive to cutoff-dependent divergences, and its ultraviolet boundary values depend on the regularisation scheme. These two definitions are exhaustive. Every local operator is either protected by Ward identities or it is not. There is no intermediate case.

3.1 Basic Principles

Under coarse-graining different operators display sharply different behaviour. The origin of this difference lies not in the details of perturbative calculations but in two definite algebraic attributes of the operator. These two attributes are direct consequences of the diagrammatic analysis carried out in Section 2. Their physical meaning can be illustrated concretely by comparing how Feynman diagrams contribute to the coarse-graining flow for different operators.

The first attribute is whether the injection term is nonvanishing. The injection term Δ_σ is the spectral weight deposited into the slow modes after the fast modes are integrated out. For a protected operator, the algebraic relation between the Noether current and the conserved charge guarantees that this term is nonvanishing at every coarse-graining step. It acts as a persistent source that drives the accumulation of spectral weight toward zero momentum. For an unprotected operator, no such guarantee exists, and the spectral function can only be diluted as coarse-graining proceeds.

The second attribute is the sign of the single-bubble contribution Π_0 . Even when the injection term is nonvanishing, whether the injected spectral weight can be amplified into a pole depends on the sign of Π_0 . This quantity is the contribution of the simplest closed fast-mode loop, the single-bubble diagram, evaluated at zero momentum. Its sign is determined entirely by the spin statistics of the fields propagating in the loop. Bosonic fields give a positive contribution, fermionic fields give a negative contribution. The sign of Π_0 controls the direction of the self-consistent feedback in the ladder resummation. When Π_0 is positive, each rung of the ladder enhances the effective coupling, and the geometric series drives the dressed polarisation toward a pole. When Π_0 is negative, each rung

suppresses the coupling, and the series remains convergent. A positive Π_0 combined with a nonvanishing injection term is therefore the structural condition for dynamical emergence through the ladder pathway; at the critical value the resummed channel forms the pole.

The combination of these two attributes yields four logical possibilities. All four are realised in the Standard Model: the energy–momentum tensor (nonvanishing injection, positive Π_0), the gauge field strength and the conserved vector current (nonvanishing injection, negative Π_0), the fermion bilinear (vanishing injection, negative Π_0), and the scalar quartic operator φ^4 (vanishing injection, positive Π_0 , the sign of the boson loop). For the zero-injection rows the sign of Π_0 does not affect the verdict: with a vanishing injection term no emergence is possible regardless of the sign.

Table 1. Operator classification by injection term and sign of Π_0 .

Injection term	Sign of Π_0	Emergence potential	Physical instance
Nonzero	Positive	Dynamical emergence possible	$T_{\mu\nu}$ (spin-2)
Nonzero	Negative	Dynamical emergence impossible	F^2, J^μ
Zero	Negative	No emergence possible	$\bar{\psi}\psi$
Zero	Positive	No emergence possible	φ^4

Note: For $T_{\mu\nu}$, the positive sign corresponds to the sign obtained on a compact internal space with curvature couplings; in flat spacetime its single-bubble contribution vanishes by the Ward identity. The classification conclusion for $T_{\mu\nu}$ is conditional on geometry activation.

This classification is rooted in the algebraic properties of the operators. For $T_{\mu\nu}$, the classification applies to the spin-2 projection; the spin-0 projection is discussed separately in Section 5.1.5. The division of labour between the protected operators originates from a fundamental difference in their coupling. The energy–momentum tensor couples to scalars and gauge fields as well as fermions, so the sign of its single-bubble contribution is not fixed by the field content alone. The conserved vector current J^μ couples only to charged fermions, and its single-bubble contribution is negative. The sign determines whether the self-consistent feedback in the ladder resummation is amplificative or suppressive.

Table 1 is a necessary consequence of conservation laws under coarse-graining. Given the conservation property of an operator and the field content to which it couples, its fate under coarse-graining is uniquely determined. The correctness of the classification does not depend on any numerical computation. Whether the injection term is nonvanishing is decided by the algebraic properties of the Noether current. The sign of Π_0 is decided by spin statistics, with the energy–momentum tensor as the sole exception: its single-bubble sign is geometry-dependent rather than fixed by the field content alone (see the note to Table 1). Both conclusions are mathematically rigorous. This logic has a structural parallel in other areas of physics. In the effective field theory treatment of large-scale structure in cosmology, the form of the effective stress tensor, with its pressure, shear viscosity and bulk viscosity, is uniquely determined by Galilean invariance and rotational symmetry [2]. Both cases embody the principle that symmetry dictates the structure of the effective theory [7].

Table 2 extends the classification to all fundamental local operators of the Standard Model. The numerical values of Π_0 quoted in the table are those of the flat-space contin-

uum computation defined in Section 5.1 (Eq. (5.1)); for the classification only the signs are relevant, the magnitudes being scheme-dependent. The division into protected and unprotected is exhaustive. Every local operator either possesses a symmetry-based protection mechanism or it does not. A noteworthy boundary case is the gluon condensate operator $F_{\mu\nu}^a F^{a\mu\nu}$. It is BRST-invariant and therefore satisfies the definition of a protected operator, yet zero-mode accumulation also requires the relevant kinetic operator to carry a protected zero-eigenspace on the background, a condition not supplied automatically for scalar-type composite channels even on compact spaces. Protection by a conservation law is necessary but not sufficient for zero-mode accumulation. The kinetic operator must also admit a non-trivial zero-eigenspace.

Table 2. Classification of fundamental Standard Model operators.

Operator / Channel	Type	Protection mechanism	CGC classification	Status
<i>Gravitational</i>				
$T_{\mu\nu}$ (spin-2)	Protected	Ward identity	0 in flat spacetime	geometry-dependent
$T_{\mu\nu}$ (spin-0)	Protected	Ward identity	0 in flat spacetime	geometry-dependent
<i>Gauge</i>				
$F_{\mu\nu}^a F^{a\mu\nu}$ (gauge field strength)	Protected	BRST symmetry	$\Pi_0 = -0.3546$	computed
G^2 (scalar glueball)	Protected	BRST invariance	$\Pi_0 = +0.1013$, zero analytic index	computed
θ -term ($F\tilde{F}$)	Protected	Topological	contact term; no pole	qualitative
<i>Currents</i>				
J^μ (conserved vector)	Protected	Ward identity	$\Pi_0 = -0.2026$	computed
$J^{5\mu}$ (axial-vector)	Unprotected	Broken chiral symmetry	zero injection	qualitative
$\bar{\psi}\gamma^\mu\psi$ (non-conserved vector)	Unprotected	No Ward identity	zero injection	logical
<i>Fermion bilinears</i>				
$\bar{\psi}\psi$ (scalar bilinear)	Unprotected	Chiral symmetry broken	zero injection	computed
$\bar{\psi}\gamma_5\psi$ (pseudoscalar bilinear)	Unprotected	Same	zero injection	logical
$\bar{\psi}\sigma^{\mu\nu}\psi$ (tensor bilinear)	Unprotected	None	zero injection	logical
<i>Higgs sector</i>				
$\ \varphi\ ^2$ (Higgs mass)	Unprotected	None	zero injection	logical
$\ \varphi\ ^4$ (Higgs quartic)	Unprotected	None	zero injection	computed
$\ D\varphi\ ^2$ (Higgs kinetic)	—	—	not a composite operator	not applicable
<i>Yukawa</i>				
$\bar{\psi}_L\varphi\psi_R$ (Yukawa coupling)	Unprotected	Chiral symmetry broken	zero injection	logical
<i>Four-Fermion</i>				
$(\bar{\psi}\psi)^2$ (four-fermion)	—	—	four-point interaction	not applicable

The status labels have the following meanings. “Computed” indicates that the Π_0 value has been evaluated via the CGC pipeline; “qualitative” that the sign is determined by spin statistics; “logical” that the classification follows from protection status; “not applicable” that the operator lies outside the framework’s scope. The enlarged table confirms the same classification pattern across all operators. Every operator that couples to a conserved charge, such as energy, momentum, electric charge or colour, falls into the protected class. Every operator that does not falls into the unprotected class. The classification is rigid. No operator straddles the boundary, and no operator changes its status under renormalisation-group evolution. For $T_{\mu\nu}$, the flat-space value is zero by the Ward identity; the nonzero sign relevant to emergence is geometry-dependent and belongs to the compact-background branch of the framework.

The gluon condensate $G^2 = F_{\mu\nu}^a F^{a\mu\nu}$ illustrates the distinction between protection and

zero-mode accumulation. Protection by BRST symmetry is necessary but not sufficient; the relevant scalar-type composite channel must also carry a protected zero-eigenspace on the compact background.

3.2 Symmetry Origin of the Criteria

The two classification criteria, the nonvanishing of the injection term and the sign of Π_0 , are not empirical or numerical properties. Both follow from symmetry, the first from the continuous symmetries of the Standard Model that give rise to Ward identities, the second from the Lorentz representation of the fields that contribute to the single-bubble diagram.

Nonvanishing injection term. The injection term is the matrix element of the operator at zero momentum transfer. A nonvanishing value is guaranteed when the operator is the Noether current of a continuous symmetry. The Ward identity $\partial_\mu J^\mu = 0$ implies, by a standard argument, that $\langle f | J^\mu(0) | i \rangle|_{q=0} \propto \langle f | Q | i \rangle$, where Q is the conserved charge. For $T_{\mu\nu}$, this charge is the total energy–momentum P_μ ; for a conserved vector current J^μ , it is the corresponding global charge. The matrix element is nonvanishing for any state carrying the corresponding quantum number.

For gauge-invariant operators such as $F_{\mu\nu}^a F^{a\mu\nu}$, BRST symmetry provides a different kind of algebraic control. The Slavnov–Taylor identities ensure that the matrix element is not forced to vanish by gauge invariance. This is weaker than the Noether current case. There is no conserved charge whose expectation value supplies a direct proof of nonvanishing. What the Slavnov–Taylor identities do supply is the absence of a mandatory suppression. Throughout this paper, the term protected for BRST-invariant operators is used in this restricted sense.

Operators with no continuous symmetry protection have no such algebraic constraint. Their injection term may be zero, or scheme-dependent; in neither case can it serve as a stable seed for ladder accumulation.

Sign of Π_0 . The sign of the single-bubble contribution is determined by spin statistics, which follows from the representation theory of the Lorentz group. Bosonic fields give positive contributions; fermionic fields give negative contributions. The sign is an algebraic fact, independent of coupling strengths, regularisation schemes, or perturbative orders.

The classification therefore rests on two independent symmetry inputs, namely the Ward/BRST structure that guarantees the injection term and the Lorentz representation that fixes the sign of Π_0 . Both are rigid because both follow from group theory, not from dynamical approximations.

Protection is necessary but not sufficient. The classification identifies the structural emergence channels. Pole formation is completed when the corresponding critical coupling is reached. Conversely, an operator may satisfy the injection and sign criteria yet be blocked by the absence of a protected zero-eigenspace, as with the scalar composite channel G^2 on the compact background. The symmetry criteria therefore select the dynamical ladder pathway, while the zero-eigenspace condition separates it from the topological branch.

Scope of the symmetry analysis. This section has considered the continuous symmetries of the Standard Model that give rise to Ward identities, namely spacetime translations, BRST symmetry, and global continuous symmetries. Discrete symmetries such as C , P , and CP do not yield Ward identities and therefore do not protect operators in the sense used here. Generalised global symmetries, while relevant to the broader structure of quantum field theory, are not needed for the classification of the fundamental local operators considered in this paper. The symmetries analysed here are the full set required for the classification; no other continuous symmetries of the Standard Model act on the local operators under consideration.

3.3 Protected Operators

Protected operators are those whose zero-momentum-transfer ladder matrix elements are forced to be nonvanishing by conservation laws under coarse-graining. Three protection sources exist.

The Ward identity originates from conserved Noether currents of spacetime translations. The energy–momentum tensor satisfies $\partial_\mu T_{\mu\nu} = 0$. Its zero-momentum matrix element equals the matrix element of the total energy–momentum P_μ , which is nonvanishing for any particle state that carries energy and momentum. The BRST symmetry protects the gauge field strength $F_{\mu\nu}^a$. It is BRST closed and its matrix elements are protected by the Slavnov–Taylor identities. The structural logic is identical to that of the energy–momentum tensor. The conservation law eliminates the mechanism that would otherwise force the matrix element to vanish at zero momentum. Noether’s theorem provides protection for conserved currents arising from global continuous symmetries. Gauge currents $J^{\mu a}$ are protected by covariant conservation, $D_\mu J^{\mu a} = 0$, which operates through the same structural logic as the ordinary Ward identity.

The proof for the energy–momentum tensor is based on the algebraic fact that the matrix element of a Noether current at zero momentum equals the matrix element of the conserved charge (2.5), which supplies a direct argument for nonvanishing. The proof for the gauge field strength is based on the same structural logic. The Ward identity provides no mandatory suppression at zero momentum transfer, a weaker but still algebraic guarantee. Zero-momentum-transfer ladder diagrams can accumulate because the injection term acts as a nonvanishing source that deposits spectral weight at every coarse-graining step. The spectral function may develop an isolated pole under coarse-graining. Whether this potential is realised depends on the dynamics of the system under the full coarse-graining flow.

3.4 Unprotected Operators

Unprotected operators are those whose zero-momentum-transfer ladder matrix elements have no conservation law protection.

The injection term of an unprotected operator lacks the algebraic constraint that forces it to be nonvanishing. For a protected operator the Ward identity or Slavnov–Taylor identity supplies an algebraic relation between the matrix element at $q = 0$ and an independently known quantity, namely the conserved charge for $T_{\mu\nu}$ or a BRST-invariant observable

for $F_{\mu\nu}^a$, and this relation directly implies that the matrix element cannot vanish. For an unprotected operator no such relation exists. The matrix element at $q = 0$ is not constrained by any symmetry, and there is no algebraic reason for it to be nonvanishing. The classification framework operates under two premises. First, coarse-graining criticality excludes fine-tuning. Second, physical observables must be scheme-independent. Given these premises, the CGC assignment for an unprotected operator is a vanishing injection term. A nonzero zero-momentum source would have to be protected by an algebraic relation; without such protection it is not stable under coarse-graining and cannot serve as a ladder seed. The framework systematically identifies which operators lack such a protected source. The zero-injection classification is therefore the structurally selected assignment under the stated premises. Neutrino mass operators, whether of Dirac type $\bar{L}\tilde{\phi}\nu_R$ or Majorana type $(L\tilde{\phi})(\tilde{\phi}^T L^c)/\Lambda$, fall into the unprotected class. Neither carries a conserved-current Ward identity. Their injection terms vanish. They cannot drive dynamical emergence through ladder resummation.

Without a nonvanishing injection term, no source exists to deposit spectral weight continuously into the slow modes. The spectral function remains continuous under coarse-graining, and no isolated pole develops. The ultraviolet boundary value of an unprotected operator is therefore not set by its own coarse-graining dynamics; it is fixed within the subtraction scheme that defines the renormalised flow. This is the subject of the following subsection.

3.5 Scheme Selection

Unprotected operators do not accumulate zero modes. Their ultraviolet boundary values are therefore not set by their own coarse-graining dynamics. The boundary condition is formulated within the subtraction scheme that removes the divergences.

Two classes of unprotected operators must be distinguished.

Take the Higgs mass parameter m^2 as an example of a positive mass-dimension operator. In a scheme that retains power-law divergences, the one-loop flow equation contains a term proportional to k^2 [19]. If one sets $m^2 = 0$ at $k = M_P$, then $\partial_t m^2 \propto M_P^2 \sim 10^{37} \text{ GeV}^2$, and the boundary condition is destroyed within the first RG step. The electroweak scale would be driven to $v \sim M_P$, deviating from the measured value by roughly seventeen orders of magnitude.

Take the Higgs quartic coupling λ as an example of a marginal operator. Even if $\lambda(M_P) = 0$ is imposed, the finite parts of the logarithmic divergences are nonvanishing at a finite cutoff. These finite parts propagate to low energies through the RG flow, making the physical value of λ scheme-dependent. Setting them to zero would amount to imposing a specific renormalisation condition, a fine-tuning under the premise that coarse-graining criticality excludes fine-tuning.

In a subtraction scheme that removes the divergences, power-law divergences are automatically set to zero by the analytic continuation of dimensional regularisation [19], and logarithmic divergences are systematically subtracted by the renormalisation procedure [20]. The boundary condition $\lambda(M_P) = 0$ can be stably maintained. Its RG stability

is guaranteed by $\partial\beta_\lambda/\partial\lambda|_{\lambda=0} < 0$, which is strictly negative over the entire error range of the input parameters [11].

The contrast between protected and unprotected operators exposes an asymmetry in scheme dependence. For protected operators the divergences have already been eliminated by Ward identities; physical results are insensitive to scheme choices. For unprotected operators the divergences survive in physical correlation functions; the correct boundary condition can only be formulated within a subtraction scheme. In the electroweak sector the Higgs quartic coupling $\|\varphi\|^4$ is an unprotected operator. Its ultraviolet boundary value is not determined by coarse-graining criticality. Within a subtraction scheme, $\lambda(M_P) = 0$ is an admissible boundary condition, and its RG stability has been verified.

3.6 Physical Consequences and Falsifiability

To test these predictions quantitatively, we have constructed the Conservation-Law-Guided Classification (CGC) pipeline. It takes a composite operator as input and returns, without relying on perturbative expansions or adjustable parameters, a verdict on its emergence potential together with the numerical value and sign of its single-bubble contribution Π_0 . The results reported below are the output of this pipeline applied to the fundamental operators of the Standard Model, spanning eight physical channels.

$T_{\mu\nu}$ is protected by the Ward identity, hence its injection term is nonvanishing. In flat spacetime the Ward identity forces its single-bubble contribution to vanish; the nonzero value relevant to emergence is geometry activated. As the spin-2 component of the only operator satisfying both emergence conditions simultaneously, it possesses the potential for dynamical emergence. If the critical condition is met, Weinberg’s low-energy theorem [21] locks the low-energy effective theory uniquely to Einstein gravity.

The spin-0 projection $P_{\mu\nu\rho\sigma}^{(0)}$ selects the trace part of $T_{\mu\nu}$. Like the spin-2 component, it vanishes in flat spacetime, and its non-zero value is geometry-dependent. The spin-0 component also possesses emergence potential and is discussed further in Section 5.1.

$F_{\mu\nu}^a F^{a\mu\nu}$ is protected by BRST symmetry, hence its injection term is nonvanishing. Its single-bubble contribution is negative, $\Pi_0 = -0.3546$. The u, d, s quark contribution dominates the total, consistent with the fermion-dominated sign structure. The injection term is nonvanishing but Π_0 is negative, so the iterative feedback in the ladder resummation is suppressive, and dynamical emergence through ladder accumulation is impossible for this channel. The zero modes of the gauge field, if they appear, must originate from a mechanism other than ladder diagram resummation, such as a topological mechanism, as the CGC framework does not address topological zero modes. The existence of gauge bosons themselves is guaranteed by gauge symmetry and is independent of this classification.

$\bar{\psi}\psi$ is unprotected, its injection term vanishes, and no emergence is possible. This negative conclusion receives independent physical support from QCD. Quark bilinears do not generate long-range correlations through ladder diagram resummation, and chiral symmetry breaking is driven by the NJL mechanism [22] rather than by spectral weight accumulation of a composite operator. The classification correctly labels this operator as unprotected and produces no false positive.

φ^4 is unprotected, its injection term vanishes, and no emergence is possible. This is consistent with the result of Ref. [11], in which the ultraviolet boundary condition $\lambda(M_P) = 0$ for the unprotected Higgs quartic coupling, combined with two-loop renormalisation-group evolution, yielded a Higgs mass consistent with the measured value within theoretical uncertainties. The classification provides the theoretical basis for that boundary condition. The ultraviolet value of an unprotected operator is not fixed by coarse-graining dynamics, and zero is the most natural choice.

Two distinct pathways to long-range correlations emerge from this analysis. The first is dynamical, proceeding through ladder resummation; the second is topological, proceeding through index theorems. The CGC framework addresses only the first; the second is identified as a separate mechanism, not analysed here.

Definite predictions follow from the classification. The injection term of any protected operator must be nonvanishing. The spectral function of a protected operator with positive Π_0 should exhibit an isolated pole. The spectral function of an unprotected operator should remain continuous. These predictions do not rely on perturbation theory or on the choice of regulator scheme. If the injection term of a protected operator were found to vanish, or if the spectral function of an unprotected operator were found to develop an isolated pole, the classification framework would be falsified. The classification framework provides explicit, falsifiable predictions that can be overturned by numerical computation.

3.7 A Hierarchical Interpretation

The classification results of Tables 1 and 2 assign a protection status to each operator. This status is fixed by the presence or absence of a Ward identity or BRST symmetry and does not change under the renormalisation-group flow. An unprotected operator has no self-sustained injection term and cannot accumulate spectral weight by itself, but this does not imply that it is dynamically inert. Its renormalisation-group evolution can still be driven by other operators. Appendix B reorganises the classification results into a hierarchical structure that exposes this driving logic. The hierarchy distinguishes operators that are self-driven from operators whose infrared dynamics is determined by others. $T_{\mu\nu}$ sits at the top as the unique self-driven operator. The protected gauge and Yukawa couplings sit at the middle level. The unprotected operators sit at the bottom. Their protection status is unchanged, but their position in the hierarchy indicates that their infrared behaviour is driven by the operators above them. A parallel branch contains G^2 , which is BRST-protected but topologically blocked. The hierarchy, summarised in Table 4 of Appendix B, offers a deeper reading of the classification, one whose driving relations can in principle be verified by functional-renormalisation-group computation.

The hierarchy also explains the ultraviolet boundary condition for the Higgs quartic coupling φ^4 . The classification has already established that its injection term vanishes. The hierarchy places φ^4 at Level 3, where its infrared dynamics is driven by the gauge and Yukawa couplings at Level 2. Its ultraviolet boundary value is therefore not set by its own dynamics. In the absence of an external scale, zero is the natural choice. It introduces no new parameters and is stable under the renormalisation-group flow. The phenomenological analysis of the Higgs sector in a companion paper adopts this boundary condition and finds

agreement with the measured Higgs mass [11]. The hierarchy thus provides the theoretical basis for that condition.

4 Irreversibility of Coarse-Graining and the RG Time

In the derivation of the Navier–Stokes equations from Boltzmann’s molecular kinetics, microscopic collision diagrams are classified by their wave number. Diagram classes not protected by conservation laws are suppressed in the thermodynamic limit. Those that are protected survive, and their resummation yields the macroscopic fluid equations [1]. The classification framework developed here follows the same sequence of classification, selection and resummation. Feynman diagrams are classified by whether the momentum transfer vanishes. Those with $q = 0$ can accumulate and those with $q \neq 0$ are suppressed. The nonvanishing of the injection term for protected operators is an algebraic consequence of the conservation laws. For protected operators with positive feedback, the resummation of ladder diagrams produces a massless spectral pole when the critical value is reached.

4.1 Irreversibility of Coarse-Graining

The classification framework established in Sections 2 and 3 determines whether a given operator possesses the potential for emergence, and along which path that emergence proceeds. The framework evaluates the algebraic and diagrammatic attributes of an operator at a given coarse-graining scale. The dynamical branch then follows the effective coupling that drives the ladder resummation and identifies the scale at which the critical value for pole formation is reached. This section examines that dynamics. The irreversibility of coarse-graining supplies an arrow of time, the RG time, which drives scalar condensation and defines a characteristic window within which the critical condition can be met.

Reference [10] established coarse-graining as the single dynamical principle governing the emergence framework. Coarse-graining is irreversible. Integrating out the fast modes removes their entanglement with the slow modes, turning the reduced density matrix of the slow modes from a pure state into a mixed state. Information is permanently lost. The irreversibility manifests itself as a definite direction of the coarse-graining flow, from the ultraviolet toward the infrared, from high energies toward low energies. The RG time $t = \ln(k/\Lambda)$ inherits this irreversibility. It flows only in one direction, and the information carried by the integrated fast modes cannot be recovered. No additional hypothesis is required. The directedness of the RG time is an intrinsic property of the Wilsonian renormalisation group [23].

This directedness of the RG time has a direct consequence for the emergence framework. The coarse-graining flow drives the system from the ultraviolet toward the infrared along a preferred direction. If the effective potential of a scalar field possesses a negative curvature at the ultraviolet scale, the RG flow will gradually reshape it. As more fast modes are integrated out, the shape of the effective potential evolves. Under appropriate conditions the curvature can change sign, and a scalar field that initially had no vacuum expectation value can acquire one. The irreversibility of coarse-graining is the dynamical origin of this process. No external time coordinate needs to be introduced.

In the effective field theory of large-scale structure in cosmology, the bias functional that describes galaxy formation is non-local in time. Galaxy formation takes a Hubble time, comparable to the time scale on which the Universe evolves, and the bias functional therefore depends on the entire formation history. In perturbation theory, however, the fields at each order evolve with a simple time dependence, and the time non-locality collapses to a set of constant coefficients [2]. The same logic applies to scalar condensation driven by the RG flow. Even if the complete real-time evolution involves non-local memory effects, under the coarse-graining flow of the RG time the microscopic history is expected to manifest itself as local and computable contributions at the macroscopic scale.

4.2 Scalar Condensation Driven by the RG Flow

The operator classification established in Section 3 identifies which operators possess the potential for dynamical emergence. For a protected operator with a positive single-bubble contribution, the ladder resummation can produce a spectral pole. Whether this potential is realised depends on the dynamics of the system under the RG flow. Scalar condensation provides a concrete setting in which the RG flow drives the system through the critical region where the conditions for emergence can be met.

The mechanism described above can be made quantitative within the functional renormalisation group. Consider a scalar field whose effective potential at the ultraviolet scale has a negative squared mass. The Wetterich equation [24] governs the evolution of the effective potential as the RG scale k is lowered. At each step the trace density, summed over the spectrum of the theory, deforms the potential. As modes are progressively integrated out, the effective potential changes shape. If the parameters are such that the curvature at the origin switches sign at some critical scale, the scalar field acquires a nonvanishing vacuum expectation value. The trajectory of the vacuum expectation value as a function of the RG scale is a pure FRG output. It contains no phenomenological parameters.

The local potential approximation provides a technically simple framework in which to carry out this computation [25]. In this approximation the spatial dependence of the scalar field is neglected. Only the homogeneous mode is retained, and the second functional derivative of the effective action reduces to the second derivative of the effective potential. The accuracy of this approximation is controlled by the anomalous dimension of the scalar field. When the anomalous dimension is small compared to the relative change of the vacuum expectation value over the relevant range of scales, the local potential approximation is reliable. The quantitative branch of the framework evaluates this condition within the FRG flow.

4.3 The Emergence Window as an RG Time Window

When scalar condensation is driven by the RG flow, the trajectory of the vacuum expectation value as a function of the RG scale defines a characteristic interval. In the early stages of the flow the vacuum expectation value is small. As the critical scale is approached it grows rapidly. Eventually it settles at its infrared equilibrium value. The interval during which the vacuum expectation value has become substantial but has not yet reached

its final value defines an RG time window. The width and position of this window are determined entirely by the FRG flow of the effective potential.

Within the emergence framework this RG time window is the dynamical arena where the emergence potential identified by the CGC classification is realised. The spectral criterion of Section 2 establishes the physical signature of emergence, an isolated massless pole in the spin two spectral density. The operator classification of Section 3 identifies which operators can produce such a pole. The RG time supplies the dynamical conditions under which the effective coupling can be driven to the critical value. Together they form the complete physical picture of emergence. The RG time provides the arrow of time for emergence.

5 Synergy between Analytic Classification and Numerical Computation

Section 4 added the dynamical dimension of coarse-graining and the arrow of time supplied by the RG time. The analytic classification and the functional renormalisation group address this dynamics from complementary directions.

The classification framework is an analytic instrument. It does not directly compute spectral functions. Instead it classifies Feynman diagrams, determining which diagram classes accumulate under coarse-graining and which are suppressed. It provides physical insight into why a spectral pole appears, which diagram classes contribute to it, and which operators are protected by conservation laws. Its task is to answer whether emergence is possible, and along which path. It does not undertake the quantitative computation of whether the effective coupling reaches the critical value. That dynamical verification is the province of the FRG.

The FRG is a numerical instrument. Starting from the complete Wetterich equation, it performs a nonperturbative computation of the effective action and outputs the full spectral function. It determines directly whether a massless pole exists in the propagator. The result is independent of the classification framework and constitutes the dynamical verification of emergence.

The two together form a complete chain from analytic discrimination to numerical confirmation. The classification framework tells the FRG what to expect. The FRG checks whether the expectation is borne out. For the energy-momentum tensor channel the analytic classification returns a verdict of dynamical emergence. The injection term is nonvanishing, and ladder diagrams can accumulate. Whether the TT propagator develops such a pole is a dynamical question, to be addressed by FRG computation in a companion paper. For the gauge field strength channel the analytic classification returns a verdict that no dynamical emergence is possible. The injection term is nonvanishing but the single-bubble contribution is negative, so the iterative feedback in the ladder resummation is suppressive. The zero modes of the gauge field, if they appear, must originate from a mechanism other than ladder diagram resummation. The FRG computation independently gives the same sign for the single-bubble contribution. The other mechanism, when present, belongs to the topological branch rather than to the CGC ladder pathway.

Further evidence for the consistency of the two approaches comes from the numerical value of the single-bubble contribution itself. The quantity Π_0 is computed along two independent code paths. One path, the classification path, uses the diagrammatic pipeline described in Section 2. The other path, the direct continuum path, computes the same quantity from the flat-space continuum bubble integrals with the same bare operator normalisation and the same field content. The two paths employ the same physical inputs but different code implementations, and both correctly apply the spin-statistics sign, with bosonic loops contributing positively and fermionic loops negatively. Both paths yield a negative Π_0 for F^2 (fermion-dominated) and a positive Π_0 for G^2 (boson-dominated), confirming the sign structure that drives the classification verdicts. This consistency provides a solid numerical foundation for the synergy between the classification framework and the FRG.

5.1 Verification on Eight Physical Channels

The verification is carried out on the operators discussed below. The numerical values reported below are obtained in the flat-space continuum limit in four spacetime dimensions, with a Gaussian cutoff e^{-p^2/Λ^2} consistent with the coarse-graining window function $W_\sigma(p) = e^{-p^2\sigma^2/2}$ used throughout this work; $\Lambda^2 = 1$ in dimensionless units regularises the ultraviolet divergence. All operators are defined with their bare normalisation, and the computation is self-contained: apart from the Standard Model field content and the electromagnetic charges Q_f , no coupling constant, mass scale or other external input enters the result. Concretely, the single-bubble contribution is

$$\Pi_0 = \sum_{\text{fields}} s_f n_f K_{\mathcal{O}} I_G(m_f^2), \quad I_G(m^2) = \frac{1}{16\pi^2} \left[1 - m^2 e^{m^2} E_1(m^2) \right], \quad (5.1)$$

where $s_f = +1$ for boson loops and -1 for fermion loops (spin statistics), n_f counts the field modes (gluons $8 \times 2 = 16$; quarks $3 \times 4 = 12$ per flavour; charged leptons 4 per flavour), and $K_{\mathcal{O}}$ is the operator kernel: $K_{\mathcal{O}} = 1$ for the Lorentz-scalar operators F^2 and G^2 , and $K_{\mathcal{O}} = Q_f^2$ for the conserved vector current $J^\mu = \sum_f Q_f \bar{\psi}_f \gamma^\mu \psi_f$, whose bubble carries the charge square of each charged fermion species. All Standard Model masses satisfy $m_f \ll \Lambda$, so that $I_G(m_f^2) = 1/16\pi^2$ in the massless limit. For the energy-momentum tensor, the Ward identity forces $\Pi_0 = 0$ in flat spacetime. The sign of Π_0 is the classification-relevant quantity; the numerical magnitude is scheme-dependent.

The results are summarised in Table 3, with explicit case studies following the pipeline sequence described in Section 2.4.

Numerical values are computed with a Gaussian cutoff e^{-p^2/Λ^2} , consistent with Ref. [10].

All logical possibilities of the classification appear among the eight channels analysed in this section, with no counterexample. In the table, “—” marks the rows that follow as logical consequences of the classification criteria rather than from direct computation. The emergence paths listed are conditional on the critical condition being reached.

Table 3. Verification matrix for the operators discussed in §5.1.

Operator	Protection source	Π_0	Dynamical emergence	Emergence path
$T_{\mu\nu}$ (spin-2)	Ward identity	0 in flat spacetime	Possible	Ladder resummation
$T_{\mu\nu}$ (spin-0)	Ward identity	0 in flat spacetime	Possible	Scalar collective mode
F^2	BRST symmetry	-0.3546	Impossible	Fermion-dominated
J^μ	Ward identity	-0.2026	Impossible	Topological, if at all
G^2 (glueball)	BRST invariance	+0.1013	Zero analytic index	None
$\bar{\psi}\psi$	None	—	Impossible	None
φ^4	None	—	Impossible	None

5.1.1 Case 1: Energy–Momentum Tensor (spin-2)

In flat spacetime the Ward identity forces the single-bubble contribution in the $T_{\mu\nu}$ channel to vanish; the nonzero value relevant to emergence is geometry activated. The transverse-traceless projector $P^{(2)}$ isolates the spin-2 channel (Case 5.1.5 treats the spin-0 projection). The ladder resummation therefore has the potential for dynamical emergence. If the critical condition is met, Weinberg’s low-energy theorem [21] locks the low-energy effective theory uniquely to Einstein gravity. The TT pole is evaluated in the companion FRG computation.

5.1.2 Case 2: Gauge Field Strength

The gauge field strength operator $F_{\mu\nu}^a F^{a\mu\nu}$ is BRST-invariant, so its injection term is non-vanishing. The single-bubble contribution, evaluated with the same continuum method, is $\Pi_0 = -0.3546$. The sign is negative because the contribution is dominated by fermion loops. The ladder resummation therefore has suppressive feedback, and dynamical emergence through ladder accumulation is impossible for this channel. The zero modes of the gauge field, if they appear, originate from a mechanism other than ladder diagram resummation and belong to the topological branch. The impossibility of ladder accumulation does not imply that gauge interactions are absent. Gauge bosons exist by virtue of gauge symmetry, independently of the CGC mechanism. The CGC analysis addresses only whether F^2 as a composite operator can form a pole through ladder resummation. The gauge field strength channel F^2 must be distinguished from the scalar glueball G^2 ; the former is the spin-1 force-carrier channel, the latter the spin-0 composite channel. Their Π_0 signs differ, negative for F^2 and positive for G^2 , and they occupy separate rows in the classification table.

5.1.3 Case 3: Fermion Bilinear

Submitted as an unprotected operator, it leads the pipeline to generate two diagrams at the one-loop level. The conservation law check confirms that no Ward identity or BRST symmetry protects the matrix elements. The injection term vanishes, and no emergence is possible. This is consistent with the NJL-driven chiral symmetry breaking discussed in Section 3.6.

5.1.4 Case 4: Higgs Quartic Coupling

Submitted as an unprotected operator, it leads the pipeline to generate four diagrams at the one-loop level. The conservation law check confirms that no symmetry protects the matrix elements. The injection term vanishes. No emergence is possible. This is consistent with the result of Ref. [11], in which the ultraviolet boundary condition $\lambda(M_P) = 0$ for the unprotected Higgs quartic coupling, combined with two-loop renormalisation-group evolution, yielded a Higgs mass consistent with the measured value within theoretical uncertainties. The classification provides the theoretical basis for that boundary condition.

5.1.5 Case 5: Energy–Momentum Tensor (spin-0)

The spin-0 projection $P_{\mu\nu\rho\sigma}^{(0)} = \frac{1}{3}\theta_{\mu\nu}\theta_{\rho\sigma}$ picks out the trace part of $T_{\mu\nu}$. As noted in Case 5.1.1, its single-bubble contribution vanishes in flat spacetime by the Ward identity, and its non-zero value is geometry-dependent. The ladder resummation therefore has the potential for dynamical emergence. If the critical condition is met, the resulting excitation is a massless scalar collective mode. Weinberg’s low-energy theorem applies only to spin-2, so the low-energy effective theory of this scalar mode is not fixed by the same argument.

5.1.6 Case 6: Conserved Vector Current

The conserved vector current $J^\mu = \sum_f Q_f \bar{\psi}_f \gamma^\mu \psi_f$ is the Noether current associated with a global continuous symmetry. Its canonical example is the electromagnetic current of the Standard Model. The operator is protected by the Ward identity $\partial_\mu J^\mu = 0$, and the zero-momentum matrix element equals the conserved charge, $\langle J^\mu(q=0) \rangle = Q$. The injection term is nonvanishing.

The single-bubble contribution is evaluated with the same continuum method of Eq. (5.1): $\Pi_0 = -\sum_f Q_f^2 n_f I_G$, the sum running over the charged fermion species to which J^μ couples, with the overall sign fixed by the spin statistics of the closed fermion loop. The result is $\Pi_0 = -0.2026$. The sign is negative because J^μ couples exclusively to charged fermions; the fermion loop dominates the spectral sum. The ladder resummation denominator remains greater than unity, and the iterative feedback is suppressive. Dynamical emergence is therefore impossible.

This channel and $T_{\mu\nu}$ share the identical protection mechanism, both being Noether currents protected by the Ward identity, yet yield opposite emergence verdicts because the fields to which they couple differ. $T_{\mu\nu}$ couples to scalars and gauge bosons as well as fermions. J^μ couples only to charged fermions. The sign of Π_0 , determined by the field content, selects the opposite outcome.

5.1.7 Case 7: Scalar Glueball (G^2)

The scalar glueball operator G^2 is the spin-0 composite of two field-strength tensors. It is distinct from the gauge field strength channel F^2 . The latter is the spin-1 propagation of gauge bosons, while the former is a bound-state channel. The scalar glueball operator $F_{\mu\nu}^a F^{a\mu\nu}$ is BRST-invariant, so its injection term is nonvanishing. The single-bubble contribution is $\Pi_0 = +0.1013$, with the positive sign from gauge-boson loops. The ladder series would therefore tend to amplify.

However, an isolated pole at $p^2 = 0$ requires not only that the ladder series diverge, but also that the relevant kinetic operator carry a protected zero-eigenspace. For scalar-type composites such as the gluon condensate, the ordinary scalar Laplacian has only the trivial constant kernel and carries no nontrivial index-protected zero mode on the compact internal space. The classification framework therefore places this channel on a topologically obstructed branch: protection and a positive Π_0 are present, but the compact spectral datum does not supply the protected zero-mode required for ladder accumulation.

The gluon condensate thus constitutes an instructive boundary case. It satisfies the protection criterion and the sign criterion, yet ladder dynamics alone does not generate the required zero-mode. The physical content is that a gluon condensate is not produced by ladder diagram resummation alone on a compact internal space. A condensate, when present, belongs instead to the effective-potential branch.

5.1.8 Case 8: Axial-Vector Current

The axial-vector current $J^{5\mu}$ is discussed in Section 3.4 as an unprotected operator. Since chiral symmetry is explicitly broken in the physical Standard Model, the injection term vanishes. It cannot develop long-range correlations under coarse-graining. This is the opposite of J^μ . The two currents have the same Lorentz structure, but different symmetry protection.

5.2 Cross-Validation Logic

The cross-validation between the classification framework and the FRG computation employs a three-category logic. When the classification returns dynamical emergence and the FRG cross-validation reports a positive sign for the single-bubble contribution, this constitutes mutual verification at the level of signs. This is the state of the energy-momentum tensor (spin-2) channel. When the classification returns dynamical emergence for a protected operator and the FRG independently confirms a positive Π_0 , this constitutes mutual verification. This is the state of the energy-momentum tensor (spin-0) channel, for which the positive sign of Π_0 is confirmed along the two independent computational paths. When the classification returns no emergence and the FRG cross-validation confirms the negative sign, this constitutes a consistent null result; this is the state of the F^2 and J^μ channels.

The cross-validation also isolates the scheme dependence of the two approaches. The analytic conclusions of the classification framework, namely the nonvanishing of the injection term, the sign of Π_0 , and the presence or absence of emergence potential, are scheme-independent results. They do not rely on the choice of regulator or on the strength of couplings. The FRG computation, performed with a specific regulator on a specific discrete spectrum, independently confirms these conclusions. The agreement between the two is a direct numerical confirmation of the analytic classification.

The FRG equilibrium flow integrated over the discrete spectrum of the compact internal space gives an enhancement of the effective coupling of order unity. The spectrum contains no hidden amplification. This result is a negative finding. It eliminates the alternative hypothesis that emergence might originate from some concealed property of the

spectrum. Had the FRG flow itself produced a large enhancement, the source of emergence could not have been precisely located. The source of emergence is cleanly isolated to a single mechanism, the Dyson self-consistent resummation. The absence of enhancement in the FRG flow acquires a clear physical meaning within the CGC analytic framework. This indicates that the emergence mechanism is isolated to the ladder resummation, not to the spectrum itself.

The hierarchical structure of operators identified in Appendix B identifies the dynamical relations evaluated by the functional-renormalisation-group branch. The driving chain from Level 1 to Level 2 to Level 3 is the RG realisation of that hierarchy.

6 Discussion and Summary

The classification rests on three diagrammatic principles derived directly from the coarse-graining operation: momentum-transfer suppression, ladder-topology accumulation, and conservation-law protection. Together with two rigid attributes of the operator—the non-vanishing of the injection term and the sign of the single-bubble contribution—these define the classification.

The classification results can be further organised into a hierarchical structure that exposes the driving relations between operators. This structure, presented in Appendix B, distinguishes self-driven operators from operators whose infrared dynamics is determined by others.

The classification criteria establish whether a given operator possesses the potential for emergence. The dynamical realisation is supplied by the nonperturbative branch of the framework. The classification framework and the functional renormalisation group perform complementary roles. The classification framework answers whether emergence is possible and along which path. The FRG evaluates the critical condition. A direct test of this synergy is provided by the numerical value of the single-bubble contribution itself. The quantity Π_0 is computed along two independent code paths, the diagrammatic pipeline and the direct continuum computation, which agree on the classification-relevant sign structure for every channel (Section 5.1). For the energy-momentum tensor channel the classification returns a verdict that dynamical emergence is possible. The injection term is nonvanishing and the single-bubble contribution, when activated by geometric curvature, is positive. The critical condition $V \cdot \Pi_0 = 4/27$ is the pole condition evaluated by the dynamical branch. The spin-0 projection of $T_{\mu\nu}$ is a particularly interesting case. When the single-bubble contribution is activated by geometric curvature, the same ladder-resummation mechanism that produces the spin-2 excitation also generates a massless scalar collective mode. The classification framework identifies this channel as a candidate for dynamical emergence. Its critical condition is part of the same FRG evaluation of the pole condition.

For the gauge field strength channel the classification returns a verdict that no dynamical emergence is possible. F^2 , with $\Pi_0 = -0.3546$, has suppressive ladder feedback because its single-bubble contribution is negative (fermion-dominated). Its zero modes, if they appear, originate from a mechanism other than ladder diagram resummation and belong to the topological branch.

Two distinct pathways to long-range correlations have emerged from this analysis. The first is dynamical, proceeding through ladder resummation of composite operators with $\Pi_0 > 0$. The second is topological, proceeding through zero modes guaranteed by index theorems on compact spaces. The CGC framework addresses only the first pathway. For F^2 , the dynamical pathway is closed ($\Pi_0 < 0$), but the topological pathway, if realised on a suitable compact space, would provide the gauge zero modes independently of ladder resummation. This distinction is essential for interpreting the classification results. The impossibility of ladder accumulation does not imply the absence of gauge interactions. Gauge bosons exist by virtue of gauge symmetry, independently of the CGC mechanism.

The cross-validation described above establishes the consistency between the analytic classification and the numerical computation at the level of static operator properties. The framework contains a dynamical dimension. The irreversibility of coarse-graining supplies the arrow of time. The RG time inherits this irreversibility and drives the condensation of scalar fields whose effective potential at the ultraviolet scale possesses a negative curvature. The trajectory of the vacuum expectation value as a function of the RG scale defines an emergence window. Inside this window the dynamical conditions for pole formation are met. Once the window is crossed, the pole forms irreversibly and is protected by the conservation laws that guarantee the nonvanishing of the injection term. The RG time structure extends the correspondence between the emergence framework and classical statistical physics from the spatial momentum level to the temporal level.

The verification on eight channels, summarised in Section 5.1, covers all logical possibilities of the classification with no counterexample. The framework has also passed benchmark tests in three known solvable physical systems. In the large- N $O(N)$ model the classification returns a verdict of dynamical emergence for the conserved Noether current, consistently with the rigorous proof that the spectral function necessarily develops a pole. In QCD the fermion bilinear is correctly classified as unprotected, producing no false positive for an operator whose chiral symmetry breaking is driven by the NJL mechanism rather than by spectral weight accumulation. In the free-field limit the sign of the single-bubble contribution reproduces the spin statistics rule.

The large- N $O(N)$ model, QCD chiral symmetry breaking and the free-field limit share none of the geometric details of the internal space that drive the emergence mechanism, yet the framework gives correct classifications in all three cases. This suggests that the core strategy, classifying Feynman diagrams by their momentum transfer and topology and then using conservation laws to determine which classes accumulate, is a general analytic tool for coarse-graining problems not merely a specific instrument of the emergence framework. Extensions to other areas, such as the fractional quantum Hall effect or strongly correlated electron systems, rely on the same diagrammatic logic, though the specific forms of the conservation laws and the relevant operators differ from one physical setting to another.

The internal consistency checks confirm that the diagram counts, the momentum transfer classification, the topology classification and the conservation law verdicts all agree with the analytic results of Appendix E of Ref. [10]. The cross-validation with two independent nonperturbative methods, the FRG and the Dyson–Schwinger equation [26], confirms the value of Π_0 and its sign structure. Whether the TT propagator develops a massless pole

is a dynamical question addressed in a companion paper. The entire chain of reasoning proceeds from the conservation laws, through the operator classification and the RG time dynamics, to the localisation of the emergence window and the independent verification of the pole. No step relies on perturbative expansions or phenomenological parameters.

The physical consequence of the $T_{\mu\nu}$ classification, namely that the low-energy effective theory is uniquely Einstein gravity if the critical condition is reached, has been established independently through a uniqueness argument based on Weinberg’s theorem [21]. The classification framework identifies the candidate operators; the theorem fixes the outcome.

A specific contribution of the CGC framework is the identification of topological obstruction as a distinct mechanism that can block zero-mode accumulation even when the dynamical conditions are satisfied. The scalar glueball G^2 is the example. Its single-bubble contribution is positive and its injection term is nonvanishing, yet it does not accumulate zero modes through the ladder pathway because the relevant scalar-type composite channel has no protected zero-eigenspace on the compact background. The framework identifies this as a topological obstruction internal to the compact spectral datum. This distinguishes protection by symmetry from accumulation of zero modes and clarifies why the two are not equivalent.

Several directions remain open.

The CGC framework has a well-defined scope. It classifies composite operators by the algebraic properties of the symmetries that protect them. It does not apply to kinetic terms in the Lagrangian, to four-point contact interactions, or to topological terms whose two-point functions are contact terms. These are boundaries of the method, not limitations of the classification logic itself.

The ladder approximation at the classification stage is used only for the qualitative judgement of whether a pole can form. This judgement is rigid. The quantitative determination of the critical scale is a task for the FRG. The FRG computation is independent of the ladder approximation and does not require a prior completeness proof. The same classification logic extends naturally to finite temperature, finite density and non-equilibrium systems. The connection between the Euclidean FRG and the Schwinger–Keldysh real-time FRG formalism supplies the corresponding real-time formulation.

The framework developed in this paper is not restricted to the emergence context. It supplies a general analytic strategy that can be applied to any quantum field theory problem involving coarse-graining and the emergence of long-range correlations. A structurally parallel logic has been rigorously established in classical statistical physics [1]. The present work brings this logic into quantum field theory.

Acknowledgments

The author declares no competing interests. This work did not involve the creation or analysis of new experimental data, and data sharing is not applicable to this article. All numerical and symbolic calculations were carried out using the CGC computational framework developed in this work and the CG-Framework nonperturbative FRG engine. The source code is available at <https://github.com/bookwormseagull1980/>

Conservation-Guided-Classification. AI-assisted language tools were used for refining the exposition and formatting of the manuscript. All scientific content, derivations, and conclusions are solely the work of the author.

A Generalised Källén–Lehmann Spectral Representation

The proof that critical slowing-down induces zero-mode accumulation in the spectral function, used in Section 2.2 as the dynamical foundation of the classification, relies on two properties of the coarse-graining operation: the monotonicity of the window function and the generalised Källén–Lehmann spectral representation. This appendix shows that both properties hold for local Hermitian operators of arbitrary spin, independently of the specific form of the interactions.

A.1 Generalised spectral representation

For any local Hermitian operator $\mathcal{O}$, the two-point correlation function in momentum space admits a spectral representation [15, 27]. For a scalar operator,

$$\langle \mathcal{O}(p) \mathcal{O}(-p) \rangle = \int_0^\infty d\mu^2 \rho_{\mathcal{O}}(\mu^2) \frac{1}{p^2 + \mu^2}, \quad (\text{A.1})$$

with $\rho_{\mathcal{O}}(\mu^2) \geq 0$ a non-negative spectral density. For a vector operator the spectral representation contains a transverse projector, the longitudinal part being eliminated by the conservation condition. For a spinor operator it contains Dirac matrix structures, but the denominator always retains the form $1/(p^2 + \mu^2)$. As in the scalar case [eqrefeq:kl-scalar](#), every two-point function can be written in the unified form

$$\langle \mathcal{O}(p) \mathcal{O}(-p) \rangle = \int_0^\infty d\mu^2 \rho_{\mathcal{O}}(\mu^2) \mathcal{P}(p) \frac{1}{p^2 + \mu^2}, \quad (\text{A.2})$$

where $\mathcal{P}(p)$ is the spin-dependent projector. For a scalar operator $\mathcal{P}(p) = 1$, for a conserved vector current $\mathcal{P}(p)$ is the transverse projector $\eta_{\mu\nu} - p_\mu p_\nu / p^2$, and for the energy–momentum tensor $\mathcal{P}(p)$ is a combination of spin-2 and spin-0 projectors.

The coarse-graining window $W_\sigma(p) = \exp(-p^2 \sigma^2 / 2)$ is a scalar function of p^2 . It commutes with the spin projector $\mathcal{P}(p)$ and acts multiplicatively on the propagator denominator. The coarse-grained two-point function is therefore written as [\(A.3\)](#)

$$\langle \mathcal{O}(p) \mathcal{O}(-p) \rangle_\sigma = \int_0^\infty d\mu^2 \rho_{\mathcal{O}}(\mu^2) \mathcal{P}(p) \frac{W_\sigma(p)}{p^2 + \mu^2}. \quad (\text{A.3})$$

The coarse-grained spectral density is defined as $\rho_{\mathcal{O}}(p^2; \sigma) = W_\sigma(p) \rho_{\mathcal{O}}(p^2)$. The arguments of the main text depend only on the denominator structure $1/(p^2 + \mu^2)$ and are independent of the specific form of the spin projector.

A.2 Monotonicity of the window function

The window function $W_\sigma(p) = \exp(-p^2 \sigma^2 / 2)$ depends monotonically on the coarse-graining scale:

$$\frac{\partial W_\sigma(p)}{\partial \sigma} = -p^2 \sigma \cdot W_\sigma(p) < 0 \quad (p^2 > 0). \quad (\text{A.4})$$

This is a mathematical property of the window function, independent of the spin or interaction type of the field. Given the non-negativity of the bare spectral function, $\rho_{\mathcal{O}}(\mu^2) \geq 0$, the coarse-grained spectral density inherits this monotonicity: $\partial \rho_{\mathcal{O}}(p^2; \sigma) / \partial \sigma \leq 0$.

A.3 Transmission of the zero-frequency divergence

The transmission of the critical slowing-down singularity to the spectral function proceeds in four steps, each independent of the operator's spin.

Step 1: Zero-frequency divergence of the order parameter. The critical slowing-down condition $\beta'(\sigma_c) = 0$ eliminates the restoring force on the order parameter. The correlation function of the order-parameter fluctuations, given in (A.5),

$$\langle \delta\sigma(\omega, \vec{p}) \delta\sigma(-\omega, -\vec{p}) \rangle = \frac{2\gamma(\sigma_c) k_B T_{\text{eff}}(\sigma_c)}{\omega^2 + [Dp^2 + (c/\sigma_c)|\beta'(\sigma_c)|]^2}, \quad (\text{A.5})$$

develops a $1/\omega^2$ divergence when $\beta'(\sigma_c) = 0$. This step depends only on the Langevin equation of the order parameter and the critical slowing-down condition. It is independent of the specific quantum fields.

Step 2: Monotonic coupling to the spectral density. The coarse-grained spectral density is $\rho_{\mathcal{O}}(p^2; \sigma) = W_{\sigma}(p) \rho_{\mathcal{O}}(p^2)$. Its dependence on σ is strictly monotonic through $W_{\sigma}(p)$.

Step 3: Transmission of fluctuations. A fluctuation $\delta\sigma$ induces a fluctuation of the window function through the chain rule, $\delta W_{\sigma} = (\partial W_{\sigma} / \partial \sigma) \delta\sigma$, which in turn induces $\delta \rho_{\mathcal{O}} = \delta W_{\sigma} \cdot \rho_{\mathcal{O}}$. By Eq. (A.4), $\partial W_{\sigma} / \partial \sigma < 0$ strictly for all $p^2 > 0$. The zero-frequency divergence of σ at $p \rightarrow 0$ is therefore transmitted without suppression to $\rho_{\mathcal{O}}$, causing its spectral weight to diverge at $p^2 = 0$.

Step 4: Identification of zero modes. The generalised Källén–Lehmann representation (A.2) implies that a divergent spectral weight at $p^2 = 0$ is equivalent to the appearance (A.6) of an isolated $\delta(p^2)$ term:

$$\rho_{\mathcal{O}}(p^2; \sigma_c) = \sum_{n \in \ker \Delta_{\mathcal{O}}} Z_n \delta(p^2) + \text{continuous part}, \quad (\text{A.6})$$

where $Z_n > 0$ are the spectral weights of the individual zero modes and $\Delta_{\mathcal{O}}$ is the kinetic operator of the field.

The above argument employs the linear-response approximation for the coupling between order-parameter fluctuations and the spectral density. A complete analysis would require a path-integral average over σ fluctuations, yielding $\langle \rho_{\mathcal{O}}(p^2; \sigma(x)) \rangle_{\sigma}$. Under critical slowing-down the dominant contribution comes from long-wavelength collective modes. The linear-response approximation captures the leading effect. Solvable benchmark models give the natural setting for checking this transmission mechanism explicitly.

A.4 Hermiticity restriction

The spectral representation (A.2) applies in its standard form to Hermitian operators [15, 27]. A non-Hermitian operator can be decomposed into Hermitian combinations $\mathcal{O}_+ =$

$\mathcal{O} + \mathcal{O}^\dagger$, $\mathcal{O}_- = i(\mathcal{O} - \mathcal{O}^\dagger)$, each of which possesses a spectral representation. The $\delta(p^2)$ isolated term is preserved under these linear combinations unless the residues of the two combinations cancel exactly. This would require fine-tuning of bare parameters, which is excluded in the emergence framework. All fundamental operators in the Standard Model are Hermitian. The hermiticity restriction does not affect the main applications of this paper.

A.5 Scope of applicability

The transmission mechanism described above assumes that the bare spectral function $\rho_{\mathcal{O}}(p^2)$ contains no power-law divergences. For protected operators this condition is guaranteed by Ward identities: conservation laws or BRST symmetry force the divergences to cancel in physical correlation functions, and the bare spectral function satisfies the required regularity conditions. For unprotected operators, power-law divergences survive in physical correlation functions. The zero-frequency divergence produced by critical slowing-down is absorbed by these cutoff-dependent divergences, and the spectral function remains continuous at the zero mode.

Section 2.2 and Section 3 together form the logical chain for the CGC ladder pathway. The dynamical mechanism establishes the mathematical route by which critical slowing-down induces zero-mode accumulation under the stated regularity and protection conditions. The operator classification selects protected positive-feedback channels as pole-forming channels at criticality, and assigns unprotected operators to the zero-injection class within the same pathway.

B Operator Hierarchy from the CGC Classification

B.1 Motivation

The CGC classification in Section 3 gives a static answer. It tells which operators have nonvanishing injection terms and which do not, but it does not tell which operators drive which, nor does it explain why some unprotected operators nevertheless acquire nontrivial infrared behaviour. This appendix reorganises the classification into a hierarchy that exposes the driving relations between operators.

B.2 Protection and Self-Driving

Protection and self-driving are distinct concepts. Protection is algebraic. It fixes whether the injection term is nonzero. A protected operator has a nonzero injection term. An unprotected operator has a vanishing one. Protection comes from Ward identities or BRST symmetry and is rigid, independent of the renormalisation-group scale. Self-driving is dynamical. It determines whether an operator can develop a spectral pole through its own dynamics alone. An operator is self-driven when its own spectral dynamics, through ladder resummation, can produce a pole independently. An operator is driven when its infrared behaviour is dominated by contributions from other operators in its renormalisation-group flow.

The two notions are independent. An unprotected operator has no self-sustained injection term and cannot accumulate spectral weight by itself, but its renormalisation-group flow can still be driven by other operators through beta functions. Protection fixes the source of the injection term. Self-driving fixes the source of the infrared dynamics.

B.3 Four Levels

Level 1: Self-Driven ($T_{\mu\nu}$). $T_{\mu\nu}$ is the Noether current of spacetime translations, protected by the Ward identity. Its injection term is nonvanishing and its infrared behaviour is set by coarse-graining criticality. It needs no other operator to drive it. The self-driven status of Level 1 rests on the spectral pole proof in Ref. [10] and on the CGC classification. Two channels emerge from Level 1: the spin-2 projection gives emergent gravity, and the spin-0 projection gives a massless scalar collective mode. Both arise from the same self-driven dynamics.

Level 2: Protected but Not Self-Driven (gauge couplings, Yukawa couplings, J^μ). The gauge couplings g_1, g_2, g_3 are protected by BRST symmetry and gauge invariance, with nonvanishing injection terms. The conserved vector current J^μ is protected by the Ward identity. The Yukawa couplings, especially y_t , are constrained by gauge symmetry. These operators cannot drive dynamical emergence through ladder resummation. For J^μ and the gauge field strength operator F^2 , the single-bubble contribution is negative because fermion loops dominate. In all cases, the positive feedback required for pole formation is absent. The gauge and Yukawa couplings thus occupy the middle level. They are protected but lack the conditions for self-driven emergence. This characterisation follows from the renormalisation-group equation structure and the absence of positive single-bubble feedback.

Level 3: Driven ($\varphi^4, \bar{\psi}\psi, \|\varphi\|^2$). These operators have vanishing injection terms and no self-sustained sources. Their evolution is driven entirely by Level 2 operators, and their ultraviolet boundary values are not set by coarse-graining criticality. The Higgs quartic coupling φ^4 is the clearest example. Its beta function is

$$\beta_\lambda = \frac{1}{16\pi^2} \left(24\lambda^2 - 6y_t^4 + \frac{3}{8} \left(2g_2^4 + \left(g_2^2 + \frac{5}{3}g_1^2 \right)^2 \right) + 4\lambda \left(3y_t^2 - \frac{9}{4}g_2^2 - \frac{17}{20}g_1^2 \right) \right) + \dots \quad (\text{B.1})$$

The y_t^4 and g^4 terms drive the renormalisation-group evolution of λ . Without them, λ would only run through its own λ^2 term. In the Standard Model, the y_t^4 term dominates. The fermion bilinear $\bar{\psi}\psi$ and the Higgs mass operator $\|\varphi\|^2$ also fall into Level 3. This characterisation follows from the beta function structure, which shows that unprotected operators are dominated by Level 2 couplings.

Level 4: Topologically Blocked (G^2). G^2 is BRST-invariant and satisfies the Level 2 protection criterion. Its injection term is nonzero and its single-bubble contribution is positive. However, its kinetic operator is a scalar Laplacian. On compact manifolds without boundary the scalar Laplacian has the trivial constant kernel but no nontrivial index-protected zero modes in the scalar composite channel (Atiyah–Singer index theorem [28];

see also Section 3.1). This is a rigid obstruction. G^2 is a separate branch. It meets the Level 2 conditions but is blocked by topology. BRST protection is necessary but not sufficient for zero-mode accumulation.

Table 4 summarises the four levels.

Table 4. Operator hierarchy from the CGC classification.

Level	Operators	Protection status	Emergence path
1	$T_{\mu\nu}$ (spin-2, spin-0)	Protected by Ward identity	Self-driven
2	Gauge, Yukawa couplings, J^μ	Protected by BRST or gauge symmetry	Protected, not self-driven
3	φ^4 , $\bar{\psi}\psi$, $\ \varphi\ ^2$	Unprotected	Driven via Level 2
4	G^2 (scalar glueball)	Protected by BRST invariance	None, topologically blocked

B.4 Structure Diagram

The hierarchy has a top-down structure. Level 1 contains $T_{\mu\nu}$, the unique self-driven operator. Level 2 contains the gauge and Yukawa couplings and the conserved vector current J^μ . They are protected but not self-driven. Level 3 contains the unprotected operators φ^4 , $\bar{\psi}\psi$, and $\|\varphi\|^2$. They are driven entirely by Level 2 operators. Level 4 is a parallel branch containing G^2 , which is BRST-protected but topologically blocked.

The ordering is logical priority, not a causal chain. The full dynamics is not one-way. Once Level 3 matter fields appear, they feed back into $T_{\mu\nu}$ through the Einstein equation. The hierarchy is an ordering of logical dependence, not a claim about temporal causality. The driving chain from Level 1 to Level 2 to Level 3 is the RG realisation of the hierarchy. Functional-renormalisation-group computation supplies the quantitative form of this flow. The hierarchy gives the structural reading of how unprotected operators inherit their infrared behaviour.

B.5 Consequences

The hierarchy leads to several consequences. $T_{\mu\nu}$ anchors the entire structure. It is self-driven and needs no external input, which explains why the gravitational channel is independent of the other interactions. The gauge and Yukawa couplings at Level 2 are protected but not self-driven. They cannot produce spectral poles through ladder resummation, and the hierarchy provides a framework for understanding where their infrared values come from. The unprotected operators at Level 3 are fully driven. Their ultraviolet boundary values are not set by coarse-graining criticality. The hierarchy fixes their logical position and identifies their boundary data as inherited from the infrared fixed-point structure of Level 2.

The Higgs quartic coupling φ^4 illustrates how the hierarchy determines ultraviolet boundary conditions for unprotected operators. Its injection term vanishes and it sits at Level 3, driven by the gauge and Yukawa couplings at Level 2. Its ultraviolet boundary value is not determined by its own dynamics. In the absence of an external scale, zero is the natural choice. The renormalisation-group stability of $\lambda = 0$ supports this identification. A companion paper adopts this boundary condition for the phenomenological analysis of the Higgs sector [11].

The topological obstruction on G^2 follows from the compact internal geometry: the scalar composite channel carries no nontrivial index-protected zero mode on the compact background (Atiyah–Singer index theorem [28]; see also Section 3.1). BRST protection is therefore necessary but not sufficient for zero-mode accumulation. This is structurally reminiscent of QCD vacuum physics, where glueball correlators are sensitive to instantons and the θ -angle. The comparison separates the compact-background obstruction from ordinary QCD glueball spectroscopy.

References

- [1] Y. Deng, Z. Hani and X. Ma, *Hilbert’s sixth problem: Derivation of fluid equations via Boltzmann’s kinetic theory*, [2503.01800](#).
- [2] M.M. Ivanov, *GGI Lectures on Large-Scale Structure Perturbation Theory (Effective Field Theory)*, [2607.28314](#).
- [3] J. McGreevy, *Generalized symmetries in condensed matter*, *Ann. Rev. Condens. Matter Phys.* **14** (2023) 57.
- [4] A.D. Sakharov, *Vacuum quantum fluctuations in curved space and the theory of gravitation*, *Sov. Phys. Dokl.* **12** (1968) 1040.
- [5] T. Jacobson, *Thermodynamics of spacetime: The Einstein equation of state*, *Phys. Rev. Lett.* **75** (1995) 1260.
- [6] E.P. Verlinde, *On the origin of gravity and the laws of Newton*, *J. High Energy Phys.* **04** (2011) 029.
- [7] D. Gaiotto, A. Kapustin, N. Seiberg and B. Willett, *Generalized global symmetries*, *J. High Energy Phys.* **02** (2015) 172.
- [8] J. Horak, F. Ihssen, J.M. Pawłowski, J. Wessely and N. Wink, *Scalar spectral functions from the spectral functional renormalization group*, *Phys. Rev. D* **110** (2024) 056009.
- [9] E. Boos and T. Ohl, *Forests and groves: Minimal gauge invariant classes of tree diagrams in gauge theories*, [hep-ph/9909487](#).
- [10] J.K. Guo, *A spectral criterion for emergent gravity*, [2607.21621](#).
- [11] J.K. Guo and J. Bai, *Ultraviolet boundary condition and the Higgs mass*, [2607.24329](#).
- [12] L. Bhardwaj, L.E. Bottini, S. Schafer-Nameki and A. Tiwari, *Categorical Landau paradigm for gapped phases*, *Phys. Rev. Lett.* **133** (2024) 161601.
- [13] X. Chen, *Generalized Landau paradigm for quantum phases and phase transitions*, [2511.19793](#).
- [14] I.G. Avramidi, *Heat kernel method and its applications*, Birkhäuser (2015), [10.1007/978-3-319-09539-6](#).
- [15] S. Weinberg, *The Quantum Theory of Fields*, vol. 1, Cambridge University Press (1995), [10.1017/CBO9781139644167](#).
- [16] D.F. Litim, *Optimized renormalization group flows*, *Phys. Rev. D* **64** (2001) 105007.
- [17] J.M. Pawłowski and M. Reichert, *Quantum gravity from the functional renormalization group*, *Front. Phys.* **8** (2020) 527.

- [18] P. Nogueira, *Automatic Feynman graph generation*, *J. Comput. Phys.* **105** (1993) 279.
- [19] G. 't Hooft and M.J.G. Veltman, *Regularization and Renormalization of Gauge Fields*, *Nucl. Phys. B* **44** (1972) 189.
- [20] S. Weinberg, *High-Energy Behavior in Quantum Field Theory*, *Phys. Rev.* **118** (1960) 838.
- [21] S. Weinberg, *Photons and gravitons in S-matrix theory: Derivation of charge conservation and equality of gravitational and inertial mass*, *Phys. Rev.* **135** (1964) B1049.
- [22] K. Fukushima and T. Hatsuda, *The phase diagram of dense QCD*, *Rept. Prog. Phys.* **74** (2011) 014001.
- [23] K.G. Wilson and J.B. Kogut, *The renormalization group and the ϵ expansion*, *Phys. Rept.* **12** (1974) 75.
- [24] C. Wetterich, *Exact evolution equation for the effective potential*, *Phys. Lett. B* **301** (1993) 90.
- [25] N. Dupuis, L. Canet, A. Eichhorn, W. Metzner, J.M. Pawłowski, M. Tissier et al., *The nonperturbative functional renormalization group and its applications*, *Phys. Rept.* **910** (2021) 1.
- [26] R. Alkofer and L. von Smekal, *The infrared behavior of QCD Green's functions*, *Phys. Rept.* **353** (2001) 281.
- [27] R.F. Streater and A.S. Wightman, *PCT, Spin and Statistics, and All That*, W. A. Benjamin (1964).
- [28] M.F. Atiyah and I.M. Singer, *The index of elliptic operators on compact manifolds*, *Bull. Amer. Math. Soc.* **69** (1963) 422.